

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Catalase {lowers the activation energy / lowers the energy needed for the reaction / provides a pathway of lower energy} (1) Any one (x1) from - Important in living cells to {increase rate of breakdown of hydrogen peroxide at {physiological / body} temperature / reactions would be too slow at body temperature (1) - Without this process hydrogen peroxide would {kill the cell / build up / be toxic to the organism} (1)	1	1		2		
	(b)	(i)		0.9 = 2 marks Award 1 mark for 0.86666667 26/30		2		2	2	2
		(ii)		labels: x axis = concentration of hydrogen peroxide / vol + y axis = rate of oxygen production / cm ³ s ⁻¹ (1) linear scales + use of at least half the graph paper (1) plots correct +/- less than one small square (1) suitable line drawn including 0 (1)		4		4	4	4

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		(iii)		- As the <u>concentration</u> of hydrogen peroxide increases, the rate (of oxygen production) increases and then levels off (1) - At lower concentrations the {hydrogen peroxide / substrate} concentration is the limiting factor / increasing concentration results in {more enzyme- substrate complexes / increased chance of successful collisions} (1) - At the higher concentrations {catalase concentration / enzyme concentration/ number of active sites} is the limiting factor / - At higher concentrations all the active sites on the catalase are full (1)	1	2		3		3
		(iv)		Any two (x1) from - Repeat experiment to {improve repeatability / reliability / to calculate a reliable mean}. (1) - Carry out the experiment with {more / different} concentrations (of hydrogen peroxide) / owtte (1) - Suggestion of a more accurate way of <u>measuring volume</u> (of oxygen), eg, a gas syringe / a burette. (1) - Control temperature using a thermostatically controlled water bath (1) - Control pH using a buffer (1)			2	2		2
		(v)		Similar shape graph but <u>plateau at a lower Vmax.</u> (1)		1		1		
				Question 3 total	2	10	2	14	6	11